

AI Job Market
2025-2026

Outputs-first dashboard. New-data processing calls the same isolated offline orchestrator used by the release pipeline.

Data source

Evidence workspace

Baseline release

Upload &
process a new dataset

Workflow

- 1. Data Basic Clean
- 2. Data Ready for ML
- 3. AI Job Market Segmentation
- 4. Model Comparison
- 5. Best Model & Importance
- 6. Salary Prediction
- 7. Integrated Market Insight
- 8. Full Pipeline

Active evidence:
FPTCranes-PR32-main

5. Best model, diagnostics and uncertainty

Saved full model · frozen-configuration CV · historically scored test evidence

Report question

How does the saved full model generalize from temporal DEV validation to the historically scored test period?

Current takeaway: Test MAE decreased by \$156 versus final-configuration DEV CV.

Evidence: Final full configuration · 298 Test rows · ui-4ee3de59fabafb88

Main limitation: R² decreased by 0.025; the Test was already exposed to scoring and is not pristine.

Saved model

Random Forest · full 13

Test R²

0.803

Test MAE

\$14,877

Test MedAE

\$4,526

Test – CV R²

-0.025

Empirical q90

±\$42,254

298 historically scored rows · period 2026-03 · evidence ui-4ee3de59fabafb88

How to read these metrics

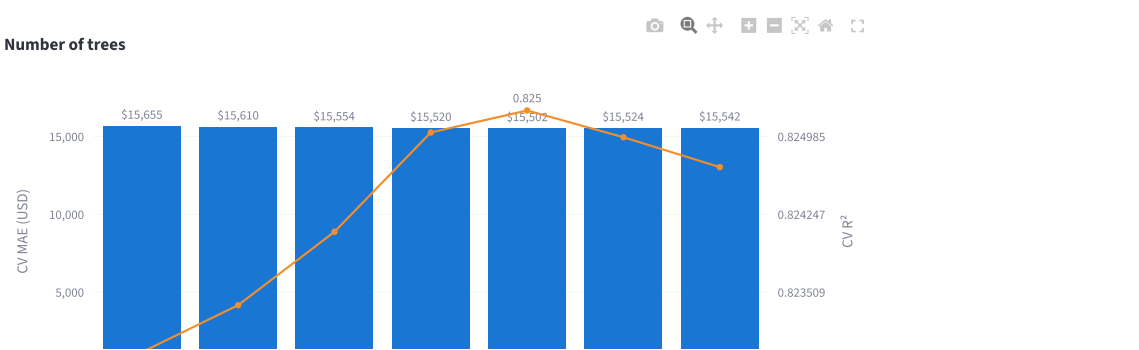
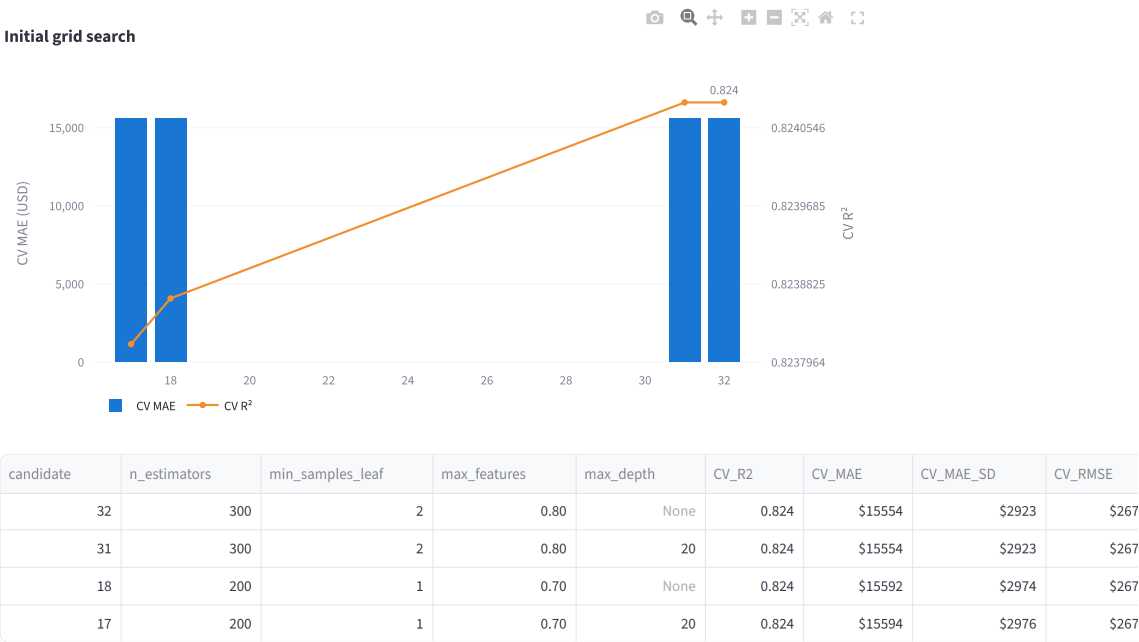
Generalization Residual diagnostics Tuning Feature reliance Trust and audit

Question: Which parameter values were applied, and what does the inherited sensitivity evidence actually support?

Branch B · How the temporal DEV folds are split

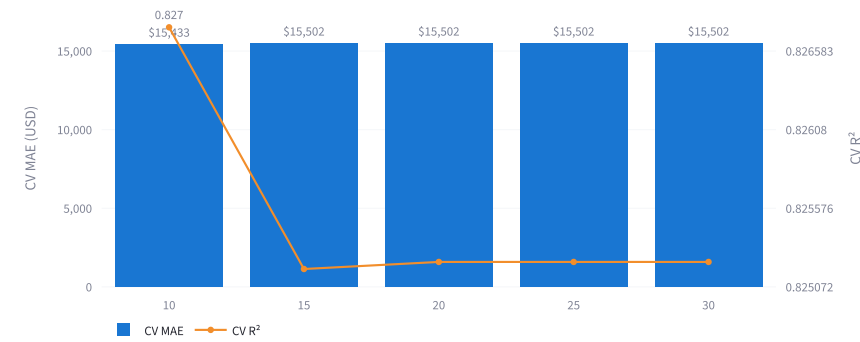
Priority guide: How hyperparameter tuning works

Initial GridSearch

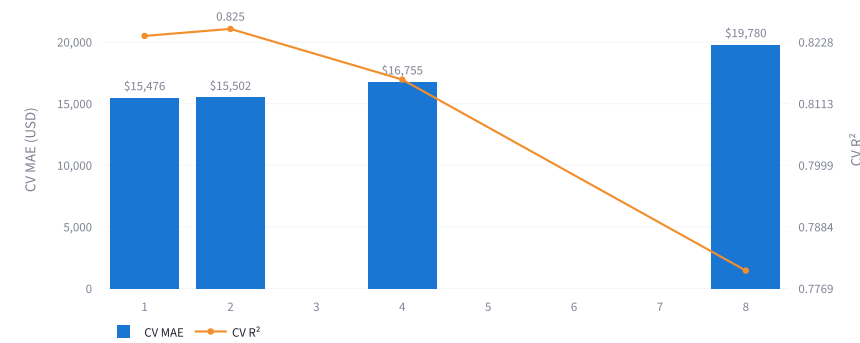




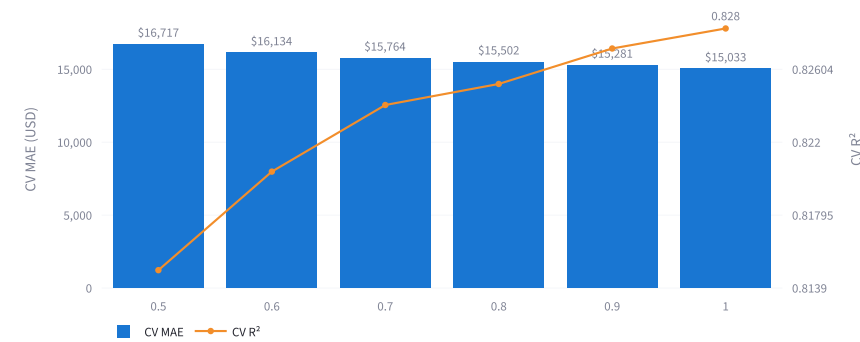
Maximum depth



Minimum leaf samples



Feature subsampling



Tuning evidence conclusion

Finding: The inherited evidence records 4 one-parameter sensitivity sweeps; the saved model applies n_estimators=400, max_depth=20, min_samples_leaf=2, max_features=1.0.

Why it matters: The displayed sweeps show local sensitivity around the inherited workflow rather than a new UI-time search.

Limit: The search was non-nested DEV evaluation ranked by R²; sweep winners do not independently establish future optimality.

Decision/use: Read applied settings from the saved model contract and use sweeps only as supporting evidence.

Bold = governing result · *Italic = reference/context*

Hyperparameter	Search space	Stage winner	Applied value	Relationship	Best CV R²	Best CV MAE
n_estimators	[200, 250, 300, 350, 400, 450, 500]	400	400	Applied = stage winner	0.825	\$15,502
max_depth	[10, 15, 20, 25, 30, None]	20	20	Applied = stage winner	0.825	\$15,502
min_samples_leaf	[1, 2, 4, 8]	2	2	Applied = stage winner	0.825	\$15,502
max_features	[0.5, 0.6, 0.7, 0.8, 0.9, 1.0]	1	1	Applied = stage winner	0.828	\$15,033

Hyperparameter	Recorded sweep	Stage winner	Best CV R ²	Best CV MAE (USD)
1. n_estimators	[200, 250, 300, 350, 400, 450, 500]	400	0.825	
2. max_depth	[10, 15, 20, 25, 30, None]	20	0.825	
3. min_samples_leaf	[1, 2, 4, 8]	2	0.825	
4. max_features	[0.5, 0.6, 0.7, 0.8, 0.9, 1.0]	1	0.828	

5/4 sensitivity charts available. Sweep winners are evidence; the model's applied parameters are the manifest contract.

Training logs

> Training log and audit downloads